

SANKETH SEQUEIRA

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Work Experience

Full Stack Developer

UIC - School of Public Health

Chicago, USA

12/2025 - Present

- Design, build, and optimize full-stack, object oriented ASP.NET Web Forms applications used by 200+ faculty, staff, and students across UIC School of Public Health.
- Work directly with deans and administrators to gather user requirements through monthly reviews and iterate on features.
- Develop frontend components using HTML, CSS, and JavaScript within ASP.NET.
- Design, modify, and maintain SQL Server schemas, and optimize queries for better performance.
- Debug production issues and improve reliability through refactoring and secure releases.

Software Engineer, Intern

KreedaLoka

Bangalore, India

07/2023 - 12/2023

- Built core systems for ChessEra used by 4 academies and 500+ users.
- Designed and developed an arena pairing system from scratch by applying matching algorithms and implemented real-time leaderboard updates using Java, C# and Unity.
- Built matchmaking and state-management logic on a WebSocket-based real-time architecture, safely handling concurrent player sessions while maintaining consistent game state and low-latency updates.
- Stored, managed, and retrieved user data for pairing using SQL, ensuring consistency and low-latency access.
- Designed modular game objects, drafted product specification documents and handled issues using JIRA.

Education

Master of Science

University of Illinois at Chicago

Chicago, USA

08/2025 - 05/2027

- Majoring in Computer Science.
- CGPA – 3.80

Bachelor of Engineering

Global Academy of Technology

Bangalore, India

12/2021 - 07/2025

- Major in Information Science and Engineering.
- CGPA – 3.61

Projects

OBDDPlus - AI-powered vehicle diagnostic platform

[GitHub](#)

12/2025

- Built a [cloud-hosted](#), microservice-based OBD-II diagnostic platform using Python and FastAPI, designed for real-time vehicle fault analysis.
- Designed and deployed two independent services (API service and AI inference service) with clear ownership boundaries, communicating over REST and orchestrated using Docker Compose.
- Deployed the system on AWS EC2, configuring Linux servers, Docker runtime, service networking, and environment-based configuration without code changes between local and cloud environments.
- Implemented defensive error handling, bounded timeouts, retry logic, and conflict-aware fix validation to generate reliable, user-friendly diagnostic explanations from overlapping vehicle sensor outputs.
- Integrated Supabase-backed storage and an agentic AI workflow with a main orchestration agent and sensor-specific subagents performing database access, LLM inference, and multi-step diagnostic reasoning.

DocBot - AI assistant for software documentation search and Q&A

[GitHub](#)

09/2025

- Built an AI-powered developer assistant using Python, FastAPI, and LLMs, exposing a REST-style API to answer natural-language questions across documentation for 10 widely used Python libraries.
- Implemented Retrieval-Augmented Generation (RAG) with vector embeddings, performing top-k similarity search and ranking to ground responses in verified source content and reduce hallucinations.
- Indexed documentation chunks using a vector similarity index and combined it with PostgreSQL for user data, query logs, and metadata, optimizing for distinct access patterns.
- Optimized performance by reusing precomputed embeddings, tuning chunk size and retrieval depth to balance latency vs response quality and avoiding full-document scans.
- Contributed to the development of a React-based web UI, implementing role-based access for admins, curators, and users.

Tool Calling Tan-Stack Table - Natural language SQL database interface

[GitHub](#)

12/2025

- React + TanStack Table UI for interacting with SQL via LLM tool calls.
- Used Cloudflare Edge (Workers) to reduce latency.
- Implemented user-facing error handling and fallback states to ensure predictable UI behavior during backend failures.

Skills

- **Programming Languages:** C# | Java | Python | C++ | C | SQL | HTML | CSS | JavaScript
- **AI & Agentic Systems:** LLMs | Agentic AI | Multi-Agent Systems | Tool Calling | LangChain | MCP | RAG | Prompt Engineering
- **Frameworks & Tools:** ASP.NET | .NET | FastAPI | Git | GitHub | Docker | Jira | VS Code | Cursor | Claude | Microsoft SQL Server
- **Cloud & DevOps:** AWS EC2 | Docker Compose | Linux | Supabase
- **Core CS:** Data Structures | Algorithms | REST APIs | Database Design | Microservices | Distributed Systems